Navrachana University

TEACHING PLAN

Data Science and Machine Learning (CMP707)

Faculty	Gauravkumarsingh Gaharwar	Division	A & B
Contact	gauravsinghg@nuv.ac.in	Office Hours	Friday: 03.00 pm to 04.00 pm
School	School of Engineering and Technology		
Program	BTech(CSE)		
Semester	Autumn	Credits	3
Academic Year	2026-27		
Lecture time & Weekdays	Wednesday (1600 Hrs – 1700 Hrs) Thursday (1000 Hrs – 1100 Hrs) Friday (1300 Hrs – 1400 Hrs)	Location	A702 A701 A701
Pre-requisites			
Course Description	This course introduces the fundamental concepts of Data Science and Machine Learning. It covers Python programming for data science, statistical methods, supervised and unsupervised learning, regression, classification, recommendation systems, and neural networks. The course focuses on applications to real-world problems and developing the skills necessary for data-driven decision-making.		
Course Abstract *			
Course Objectives	1. Introduce the field of Data Science and its applications. 2. Build a strong foundation in statistics and data visualization. 3. Teach core supervised and unsupervised machine learning methods. 4. Explore modern techniques in deep learning and recommendation systems. 5. Enable students to build end-to-end data science projects.		
Learning Outcomes	*After completing this course, students will be able to* *CO1. Apply Python and statistical techniques to real-world data.* *CO2. Build and validate machine learning models.* *CO3. Interpret model results and apply them to decision-making.* *CO4. Understand and implement recommendation systems and neural networks.*		
Typology of Course	Theory / Tutorial		
Course Outline (Units, Hours, Textbooks, Reference Books)	**Unit 1: Introduction to Data Science and Python:** Introduction to Data Science, Python Basics, Numpy, Pandas, Data Visualization **Unit 2: Statistics and Probability for Data Science:** Descriptive and Inferential Statistics, Probability Distributions, Hypothesis Testing **Unit 3: Supervised Learning:** Regression, Classification, Logistic Regression, KNN, SVM, Evaluation Metrics **Unit 4: Unsupervised Learning:** Clustering, PCA, Dimensionality Reduction, Anomaly Detection		

	Unit 5: Recommendation Systems and Deep Learning: Collaborative Filtering, Neural Networks, CNNs, Applications
Pedagogy	Explaining each topic with an example and then allow students to solve similar types of problems
Expectations from Students *	Students should attend 100% of classes Timely submission of assignments
Assessment / Evaluation	Class Tests, Surprise Test, Research Paper, Presentation, Class Participation
Attendance Policy	The University expects 100% attendance, but a minimum of 80% attendance is mandatory in each course to be eligible to appear for the end-semester examination of the course. a) Dean/Head of the School can recommend the attendance upto 15% in the course or courses for representing the university at any regional, national or international competition in the field of academics or sports or due to long duration ill-health or other emergency situations. Provost based on the case submitted by Registrar's Office with recommendation of the Dean/School head will take decision. b) For beyond 15% in any genuine cases, the decision to condone the attendance will be taken by the Provost on merits of the case. c) Documentary evidence is required to receive excusable absence and should be submitted to the Provost with the recommendation of the Dean/Head of the School. d) A student not meeting the attendance requirements in a course will not be allowed to appear for the regular end semester examination and will be awarded 'F' grade in that course. However, she/he may be permitted for re-examination. e) A student with very low attendance i.e. less than 65% in all courses during the current semester may not be allowed to appear for regular end semester examination. She/he will have to re-register all courses during the next academic year. f) Students playing individual sports or team sports at regional/national/international level with authorized sports body of the respective games will be exempted from attendance during match/competition days as well as during practice session/camps on producing relevant documents.
Project / Assignment Details *	
Course Material	**Reference Books:** 1. Joel Grus, "Data Science from Scratch", O'Reilly Media. 2. Aurélien Géron, "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow", O'Reilly Media. 3. Trevor Hastie, Robert Tibshirani, and Jerome Friedman, "The Elements of Statistical Learning". 4. Ian Goodfellow, Yoshua Bengio, and Aaron Courville, "Deep Learning", MIT Press.
Additional Information *	

* These are optional fields.

Session Plan

Topic Title	Session No.	Topic & Subtopic Details	Readings, Cases, etc.	Activities*	Important Dates
Unit-I **Introduction to Data Science and Python**	L-1	Introduction to Data Science	Textbook – 1		
	L-2	Introduction to Data Science	Textbook – 1		
	L-3	Python Basics	Textbook – 1		
	L-4	Python Basics	Textbook – 1		
	L-5	Numpy	Textbook - 1		
	L-6	Numpy	Textbook – 1		
	L-7	Pandas	Textbook – 1		
	L-8	Pandas	Textbook - 1		
	L-9	Data Visualization	Textbook - 1		
Unit-II **Statistics and Probability for Data Science**	L-10	Descriptive and Inferential Statistics	Textbook - 3		
	L-11	Descriptive and Inferential Statistics	Textbook - 3		
	L-12	Descriptive and Inferential Statistics	Textbook - 3		
	L-13	Probability Distributions	Textbook - 3		
	L-14	Probability Distributions	Textbook - 3		
	L-15	Probability Distributions	Textbook - 3		
	L-16	Hypothesis Testing	Textbook - 3		
	L-17	Hypothesis Testing	Textbook - 3		
	L-18	Hypothesis Testing	Textbook - 3		
Unit-III **Supervised Learning**	L-19	Regression	Textbook - 2		
	L-20	Regression	Textbook - 2		
	L-21	Classification	Textbook - 2		
	L-22	Classification	Textbook - 2		
	L-23	Logistic Regression	Textbook - 2		
	L-24	KNN	Textbook - 2		
	L-25	SVM	Textbook - 2		
	L-26	Evaluation Metrics	Textbook - 2		
	L-27	Evaluation Metrics	Textbook - 2		
Unit-IV **Unsupervised Learning**	L-28	Clustering	Textbook - 2		
	L-29	Clustering	Textbook - 2		
	L-30	PCA	Textbook - 2		
	L-31	PCA	Textbook - 2		
	L-32	Dimensionality Reduction	Textbook - 2		
	L-33	Dimensionality Reduction	Textbook - 2		
	L-34	Anomaly Detection	Textbook - 2		
	L-35	Anomaly Detection	Textbook - 2		
	L-36	Anomaly Detection	Textbook - 2		
Unit-V **Recommendation Systems and Deep Learning**	L-37	Collaborative Filtering	Textbook - 4		
	L-38	Collaborative Filtering	Textbook - 4		
	L-39	Neural Networks	Textbook - 4		
	L-40	Neural Networks	Textbook - 4		
	L-41	CNNs	Textbook - 4		
	L-42	CNNs	Textbook - 4		
	L-43	Applications	Textbook - 4		
	L-44	Applications	Textbook - 4		
	L-45	Applications	Textbook - 4		